

SIMATS ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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OS Managing Process Scheduling, Memory Allocation, And I/O Operations In A Modern Compute

CSA0407-OPERATING SYSTEM

Guided By:
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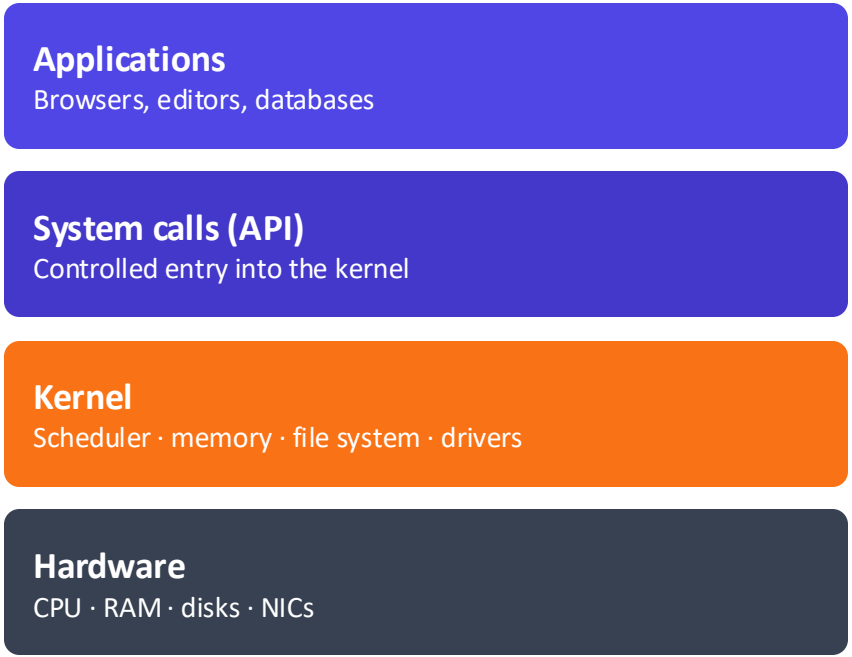
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What an operating system does

Software that manages hardware and provides services to applications.

Core responsibilities

- Resource management: CPU, memory, storage, devices
- Abstraction: files, processes and sockets instead of raw hardware
- Isolation and protection between users and programs
- Services: system calls, drivers, networking, UI
- Fair and efficient sharing under concurrent load



Kernel architectures

Where the line between kernel and user space is drawn.

Monolithic

Whole OS runs in kernel mode

- Fast: direct function calls
- Large trusted code base
- A driver bug can crash the system

Examples: Linux, classic UNIX

Microkernel

Only core services in the kernel

- Servers run in user space
- Better fault isolation
- IPC overhead costs speed

Examples: QNX, seL4, MINIX

Hybrid

Mix of both approaches

- Performance-critical code in kernel
- Modular subsystems
- Compromise on size and speed

Examples: Windows NT, macOS (XNU)

Processes, concurrency and deadlock

Running many activities at once without corrupting shared state.

Process vs thread

- Process: own address space and resources
- Thread: lightweight unit sharing its process memory
- Context switch is cheaper between threads

Synchronization

- Race conditions on shared data
- Mutexes, semaphores, monitors, atomics

Deadlock needs all four (Coffman)

1. Mutual exclusion
2. Hold and wait
3. No preemption
4. Circular wait

Break any one condition to prevent deadlock, e.g. impose a global lock ordering to avoid circular wait.

Memory and file systems

Virtual memory for isolation, file systems for persistence.

Virtual memory

- Each process sees a private address space
- Paging: page tables map pages to frames; TLB caches translations
- Page faults load data on demand from disk
- Replacement: LRU approximations, clock
- Thrashing when working sets exceed RAM

File systems

- Hierarchy of directories and files
- Inodes store metadata and block pointers
- Journaling protects consistency after a crash
- Permissions and ACLs control access
- Examples: ext4, NTFS, APFS, XFS

Major operating systems compared

Different design goals lead to different strengths.

System	Kernel	Strengths	Typical use
Windows	NT (hybrid)	Broad hardware and software support; enterprise management	Desktops, business, gaming
Linux	Monolithic, modular	Open source, stable, highly customizable	Servers, cloud, embedded
macOS	XNU (hybrid)	Tight hardware integration, UNIX base	Creative and developer desktops
Android	Linux-based	Touch UI, app sandboxing, power management	Phones, tablets, wearables

THANK YOU


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